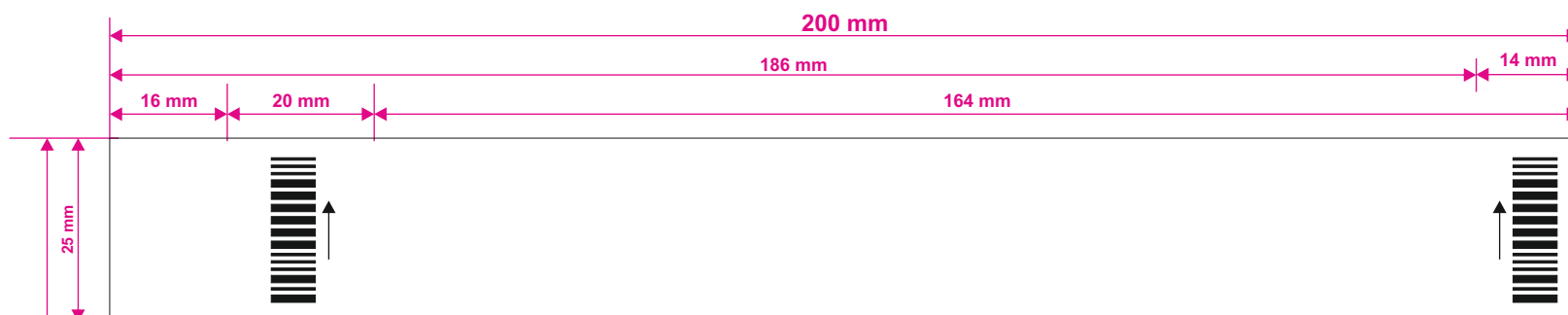


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For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory Clopi-75 Clopidogrel Tablets USP 75 mg Composition Each film coated tablet contains: Clopidogrel Bisulfate USP equivalent to Clopidogrel 75 mg Description: Pink coloured, circular shaped, biconvex, film coated tablets debossed "L 11" on one side and plain on other side. Indications Prevention of atherothrombotic events in Patients suffering from MI (from a few days until <35 days), ischemic stroke (from 7 days until <6 months) or established peripheral arterial disease. Patients suffering from Non-ST segment elevation acute coronary syndrome (unstable angina or non-Q-wave MI), including patients undergoing stent placement following percutaneous coronary intervention; ST segment elevation acute MI, in combination with Acetylsalicylic acid in medically treated patients eligible for thrombolytic therapy. Pharmacology Mechanism of Action Clopidogrel is an inhibitor of platelet aggregation. A variety of drugs that inhibit platelet function have been shown to decrease morbid events in people with established atherosclerotic cardiovascular disease as evidenced by stroke or transient ischemic attacks, myocardial infarction, or need for bypass or angioplasty. This indicates that platelets participate in the initiation and/or evolution of these events and that inhibiting them can reduce the event rate. Pharmacodynamic Effect Clopidogrel selectively inhibits the binding of adenosine diphosphate (ADP) to its platelet receptor and the subsequent ADP-mediated activation of the glycoprotein GPIIb/IIIa complex, thereby inhibiting platelet aggregation. Biotransformation of Clopidogrel is necessary to produce inhibition of platelet aggregation, but an active metabolite responsible for the activity of the drug has not been isolated. Clopidogrel also inhibits platelet aggregation induced by agonists other than ADP by blocking the amplification of platelet activation by released ADP. Clopidogrel does not inhibit phosphodiesterase activity. Clopidogrel acts by irreversibly modifying the platelet ADP receptor. Consequently, platelets exposed to Clopidogrel are affected for the remainder of their lifespan. Dose dependent inhibition of platelet aggregation can be seen 2 hours after single oral doses of Clopidogrel. Repeated doses of 75mg Clopidogrel per day inhibit ADP-induced platelet aggregation on the first day, and inhibition reaches steady state between Day 3 and Day 7. At steady state, the average inhibition level observed with a dose of 75mg Clopidogrel per day was between 40% and 60%. Platelet aggregation and bleeding time gradually return to baseline values after treatment is discontinued, generally in about 5 days. Pharmacokinetics Absorption and Distribution: Clopidogrel is rapidly absorbed after oral administration of repeated doses of 75mg Clopidogrel (base), with peak plasma levels (3 mg/L) of the main circulating metabolite occurring approximately 1 hour after dosing. The pharmacokinetics of the main circulating metabolite are linear (plasma concentrations increased in proportion to dose) in the dose range of 50 to 150mg of Clopidogrel. Absorption is at least 50% based on urinary excretion of Clopidogrel related metabolites. Clopidogrel and the main circulating metabolite bind reversibly in vitro to human plasma proteins (98% and 94%, respectively). The binding is nonsaturable in vitro up to a concentration of 100 µg/ml.	Metabolism and Elimination: <i>In vitro</i> and <i>in vivo</i>, Clopidogrel undergoes rapid hydrolysis into its carboxylic acid derivative. In plasma and urine, the glucuronide of the carboxylic acid derivative is also observed. The oxidative step is regulated primarily by Cytochrome P450 ISOENZYMES 2B6, 3A4, 1A1, 1A2 and 2C19. Dosage and Administration 1 tablet daily. Non-ST segment elevation acute coronary syndrome (unstable angina or non-Q wave MI) Initially 300 mg as a single loading dose & continued at 75 mg once a day (with Acetylsalicylic acid 75-325 mg daily). ST segment elevation acute MI Single daily dose of 75 mg, initiated w/ a loading dose in combination with Acetylsalicylic acid & with or without thrombolytics. Elderly >75 yr Initiate without loading dose. Combination therapy should be started immediately after symptoms start & continued for at least 4 weeks. Overdosage Overdose of clopidogrel may lead to prolonged bleeding time and subsequent bleeding complications. No antidote to the pharmacological activity of clopidogrel has been found. If prompt correction of prolonged bleeding time is required, platelet transfusion may reverse the effects of clopidogrel. Contraindications The use of Clopidogrel is contraindicated in the following conditions: Hypersensitivity to the drug substance or any component of the product. Active pathological bleeding such as peptic ulcer & intracranial haemorrhage, severe liver impairment, Lactation. Special warning & Precautions for use: Pharmacogenetics: Based on literature data, patients with genetically reduced CYP2C19 function (intermediate or poor metabolisers) have lower systemic exposure to the active metabolite of clopidogrel and diminished antiplatelet responses, and generally exhibit higher cardiovascular event rates following myocardial infarction than do patients with normal CYP2C19 function. Warning Reduced effectiveness due to impaired CYP2C19 function: The inhibition of platelet aggregation by clopidogrel is entirely due to an active metabolite. Clopidogrel is metabolized to this active metabolite in part by CYP2C19. This metabolism can be impaired by genetic variations in CYP2C19 and by concomitant medications that interfere with CYP2C19. Avoid use of Clopidogrel in patients with impaired CYP2C19 function due to known genetic variation or due to drugs that inhibit CYP2C19 activity. Omeprazole: Co-administration of Clopidogrel with omeprazole, a proton pump inhibitor that is an inhibitor of CYP2C19, reduces the pharmacological activity of Clopidogrel, if given concomitantly or if given 12 hours apart. Thrombotic thrombocytopenic purpura (TTP): TTP has been reported rarely following use of Clopidogrel, sometimes after a short exposure (<2 weeks). TTP is a serious condition that can be fatal and requires urgent treatment including plasmapheresis (plasma exchange). It is characterized by thrombocytopenia, microangiopathic hemolytic anemia (schistocytes [fragmented RBCs] seen on peripheral smear), neurological findings, renal dysfunction, and fever. Precaution General: Clopidogrel prolongs the bleeding time and therefore should be used with caution in patients who may be at risk of increased bleeding from trauma, surgery, or other pathological conditions (particularly gastrointestinal and intraocular). If a patient is to undergo elective surgery and an antiplatelet effect is not desired, Clopidogrel should be discontinued 7 days prior to surgery. Due to the risk of bleeding and undesirable hematological effects, blood cell count determination and/or other appropriate testing should be promptly considered, whenever such suspected clinical symptoms arise during the course of treatment. Use in Hepatically-Impaired Patients: Experience is limited in patients with severe hepatic disease, who may have bleeding diatheses. Clopidogrel should be used with caution in this population.

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200 mm

Carcinogenesis, Mutagenesis, Impairment of Fertility: There was no evidence of tumorigenicity when clopidogrel was administered for 78 weeks to mice and 104 weeks to rats at dosages up to 77 mg/kg per day, which afforded plasma exposures >25 times that in humans at the recommended daily dose of 75 mg.

Clonidogrel was not genotoxic in in vitro tests and in vivo test

Clopidogrel was found to have no effect on fertility of male and female rats at oral doses up to 400 mg/kg per day (52 times the recommended human dose on a mg/m² basis).

Pediatric use: Safety and effectiveness in the pediatric population have not been established.

Interactions

Clopidogrel is metabolized to its active metabolite in part by CYP2C19. Concomitant use of drugs that inhibit the activity of this enzyme results in reduced plasma concentrations of the active metabolite of clopidogrel and a reduction in platelet inhibition. Avoid concomitant use of drugs that inhibit CYP2C19, including omeprazole, esomeprazole, cimetidine, fluconazole, ketoconazole, voriconazole, etravirine, felbamate, fluoxetine, fluvoxamine, and ticlopidine.

Aspirin: Aspirin did not modify the clopidogrel-mediated inhibition of ADP-induced platelet aggregation. Clopidogrel potentiated the effect of aspirin on collagen-induced platelet aggregation. Clopidogrel and aspirin have been administered together for up to one year.

Heparin: Coadministration of heparin had no effect on inhibition of platelet aggregation induced by Clopidogrel.

Nonsteroidal Anti-Inflammatory Drugs (NSAIDs): The concomitant administration of Clopidogrel was associated with increased occult gastrointestinal blood loss. NSAIDs and Clopidogrel should be coadministered with caution.

Warfarin: Because of the increased risk of bleeding, the concomitant administration of warfarin with Clopidogrel is not recommended.

Other Concomitant Therapy: No clinically significant pharmacodynamic interactions were observed when Clopidogrel was coadministered with atenolol, nifedipine, or both atenolol and nifedipine. The pharmacodynamic activity of Clopidogrel was also not significantly influenced by the co-administration of phenobarbital, cimetidine or estrogen.

The pharmacokinetics of digoxin or theophylline were not modified by the co-administration of Glipidoorel bisulfate.

Since clopidogrel is metabolized to its active metabolite by CYP2C19, use of drugs that inhibit the activity of this enzyme would be expected to result in reduced drug levels of the active metabolite of clopidogrel and a reduction in clinical efficacy. Concomitant use of drugs that inhibit CYP2C19 (eg, Proton pump inhibitors) should be discouraged.

Side Effects

1. Blood and lymphatic system disorders: Agranulocytosis, aplastic anemia/pancytopenia, thrombotic thrombocytopenic purpura (TTP).
2. Gastrointestinal disorders: Gastrointestinal and retroperitoneal hemorrhage with fatal outcome, colitis (including ulcerative or lymphocytic colitis), pancreatitis, stomatitis.
3. General disorders and administration site condition: Fever, hemorrhage of operative wound
4. Hepato-biliary disorders: Acute liver failure, hepatitis (non-infectious), abnormal liver function test
5. Immune system disorders: Hypersensitivity reactions, anaphylactoid reactions, serum sickness
6. Musculoskeletal, connective tissue and bone disorders: Musculoskeletal bleeding, myalgia, arthralgia, arthritis
- 7 Nervous system disorders: Taste disorders, fatal intracranial bleeding
- 8 Eye disorders: Eye (conjunctival, ocular, retinal) bleeding.
9. Psychiatric disorders: Confusion, hallucinations
10. Respiratory, thoracic and mediastinal disorders: Bronchospasm, interstitial pneumonitis, respiratory tract bleeding
11. Renal and urinary disorders: Glomerulopathy, increased creatinine levels
- 12.Skin and subcutaneous tissue disorders: Maculopapular or erythematous rash, urticaria, bullous dermatitis, eczema, toxic epidermal necrolysis, Stevens-Johnson syndrome, angioedema, erythema multiforme, skin bleeding, lichen planus
- 13.Vascular disorders: Vasculitis, hypotension

Pregnancy and Lactation

Pregnancy category B: Reproduction studies performed in rats and rabbits at doses up to 500 and 300 mg/kg/day (respectively, 65 and 78 times the recommended daily human dose on a mg/m² basis), revealed no evidence of impaired fertility or fetotoxicity due to clopidogrel. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of a human response, Clopidogrel should be used during pregnancy only if clearly needed.

Nursing mothers: It is not known whether this drug is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the nursing woman.

Effect on ability to drive and use machine:

No effect on the ability to drive or operate complicated machinery.

Storage

Storage
Store below 30°C.

Presentation

Blister pack of 10 Tablets.

Manufactured by:

Manufactured by:
Macleods Pharmaceuticals Limited,
Block N-2, Village Theda, Post Office Lodhimajra, Tehsil Baddi,
District Solan, Himachal Pradesh-174101, India.

Product Registration Holder:

Product Registration Holder:
Macleods Pharmaceutical Sdn. Bhd.,
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Date of revision:

Date of revision.
24th December 2024